PROJECT SYNOPSIS

Project Title: Dynamic Schema Data Collection & Analytics Engine (D-SCAE)

Team Members:

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1. Abstract

The **Dynamic Schema Data Collection & Analytics Engine (D-SCAE)** is a web-based enterprise application designed to facilitate the rapid creation, distribution, and analysis of structured data collection events (surveys, registrations, and polls).

Unlike traditional database applications where the schema is static (fixed columns), D-SCAE utilizes a **Meta-Data Driven Architecture**. This allows administrators to define custom data fields, validation rules, and participation criteria at runtime without altering the underlying database structure. The system solves the "Schema Rigidity" problem by implementing a hybrid relational model capable of storing heterogeneous data types while enforcing strict integrity constraints.

2. Problem Statement

Organizations frequently need to collect structured information from specific subsets of users (e.g., "Feedback from 4th Sem CSE Students" or "Registration for Hackathon"). Standard implementations require creating new database tables for every new form, which is inefficient and unscalable. Existing solutions often lack granular, attribute-based access control (e.g., restricting a form only to users who meet specific criteria).

3. Objectives & Key Features

- **Dynamic Form Generation:** Administrators can create data collection events with user-defined attributes. The system supports multiple data types (Text, Numeric, Boolean, Date) and enforces constraints (Required, Range, Regex Pattern) defined at creation time.
- **Rule-Based Access Control (RBAC):** The system implements a predicate-based eligibility engine. Administrators can define logical rules (e.g., Branch = 'CSE' AND CGPA > 8.0) to dynamically determine which users are authorized to view and submit a specific form.
- **Temporal Constraints & Automation:** Events are bound by strict deadlines. The system includes an automated scheduler to close submissions upon deadline expiry and trigger reminder notifications to eligible users who have not yet responded.
- **Statistical Aggregation & Analytics:** A built-in analytics module generates real-time summaries of the collected data. Administrators can view public or private statistics (e.g., participation rates, average scores, frequency distributions) derived from the dynamic datasets.

4. Methodology

The project will be implemented using a **Relational Database Management System (RDBMS)**. To handle the dynamic nature of user-defined fields, the database will employ an **Entity-Attribute-Value (EAV)** model or **JSON-based storage** within the relational schema.

- **Backend Logic: Stored Procedures** will be used to validate incoming data against the meta-defined rules (e.g., checking if a user's input falls within the allowed numeric range).
- **Conflict Resolution: Triggers** will enforce logical constraints, such as preventing duplicate submissions or submissions past the deadline.

5. Tools & Technologies

- **Database:** MySQL / Oracle (PL/SQL for business logic)
- **Backend Framework:** Java (JSP/Servlet) or Python (Flask/Django) or Node.js
- **Frontend:** HTML5, CSS3, JavaScript (for dynamic form rendering)
- **Development Environment:** Eclipse / VS Code / MySQL Workbench